

ASSIGNMENT #01

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Name : Lama Alsurakhi

Due Date: Friday, January 16th TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

CHE 301 Assignment #01
Ideal Brayton Cycle and Numerical Methods

Lama Alsurakhi

Spring 2026

Abstract

This is a complete report of the solutions to Assignment #1 for CHE 301: Chemical Engineering Thermodynamics. In Problem A it discusses “classical thermodynamics” by the analysis of the Ideal Brayton Cycle while operating under steady state conditions. The compressor pressure ratio is derived analytically and by using the Excel solver to verify that it maximizes the net power output. All equations, the simplifications of the equations and all the substitutions are shown. All the thermodynamics properties such as efficiencies, net power, and the minimum air flow rate required to produce MW are calculated. Problem B focuses on getting the numerical solutions of the nonlinear equations for the vapor fractions using three methods in excel, the Newton’s Method, the Secant Method, and the Goal Seek Function, with all iterations described and images of the charts are added.

Contents

1	Background	4
2	Problem A: Review of Classical Thermodynamics. Ideal Brayton Cycle	4
2.1	Given Data and Assumptions	4
2.2	Derivation of Optimal Pressure Ratio	5
2.2.1	Derivation of Isentropic Relations Starting from $PV^\gamma = C$	5
2.2.2	General Steady-State Energy Balance	7
2.2.3	Energy Balance for the Compressor	7
2.2.4	Energy Balance for the Turbine	8
2.2.5	Net Work Expression	8
2.2.6	Taking the Derivative	9
2.3	Excel Solver Method	9
2.4	Finding T_2 , T_4 , and P_2 :	10
2.5	Finding the thermal efficiency:	11
2.6	Finding the Carnot Efficiency	12
2.7	Calculating net power at maximum efficiency:	12
2.8	Minimum Air Flow Rate:	12
3	Problem B: Numerical Solution of an Algebraic Equation	13
3.1	Finding the Derivative:	13
3.2	Calculating the Vapor fraction α :	13
3.2.1	By Using Newton method:	14
3.2.2	By using secant method :	14
3.3	By Using Goal seek:	15
4	Conclusion	16

1 Background

The Brayton Cycle is a thermodynamic cycle that is used to model gas. The cycle consists of a compressor, a combustor, and turbine all operating together at steady state conditions. When the element in the system is assumed to behave as ideal gas, then constant heat capacity and isentropic compression and expansion are assumed.

Numerical methods such as Newton's Method, Secant Method, and the Goal Seek Function in Excel are most commonly used especially in chemical engineering to solve nonlinear equations that are in equilibrium. Those methods are most commonly used for vapor-liquid equilibrium calculations. These methods provide accurate solutions when analytical solutions are not possible.

2 Problem A: Review of Classical Thermodynamics. Ideal Brayton Cycle

2.1 Given Data and Assumptions

Air enters a compressor at a pressure of $P_1 = 100 \text{ kPa}$ and a temperature of $T_1 = 300 \text{ K}$. The air is compressed adiabatically and isentropically to a pressure P_2 before entering the combustor. The combustor products reach a temperature of $T_3 = 1400 \text{ K}$ and are expanded adiabatically and isentropically in the turbine to a pressure $P_4 = P_1$.

The following assumptions apply:

- Steady-state operation
- Ideal gas behavior
- Constant molar heat capacity
- Equal molar flow rate through all components
- Negligible kinetic and potential energy changes
- Adiabatic and isentropic compressor and turbine

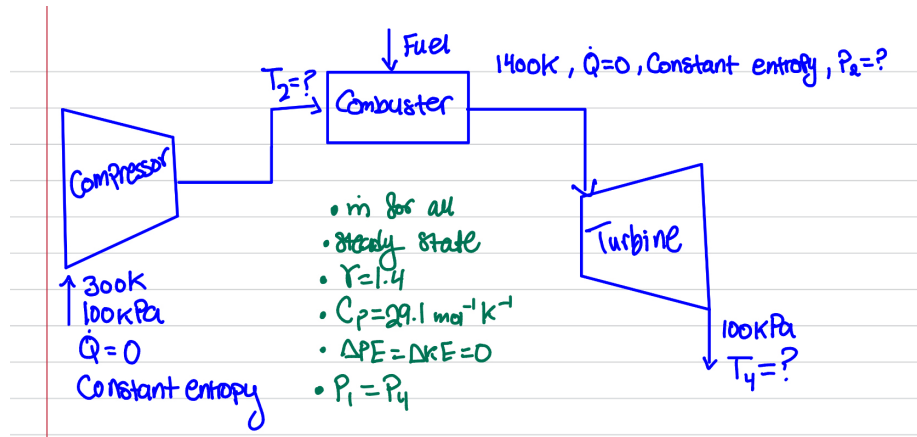


Figure 1: Hand Drawn Image of the System

2.2 Derivation of Optimal Pressure Ratio

2.2.1 Derivation of Isentropic Relations Starting from $PV^\gamma = C$

For an isentropic process involving an ideal gas, the pressure and specific volume are related by:

$$PV^\gamma = C \quad (1)$$

where C is a constant.

Consider two thermodynamic states, state 1 and state 2:

$$P_1 V_1^\gamma = P_2 V_2^\gamma \quad (2)$$

Rearranging:

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma \quad (3)$$

Taking both sides to the power $1/\gamma$:

$$\left(\frac{P_2}{P_1} \right)^{1/\gamma} = \frac{V_1}{V_2} \quad (4)$$

For an ideal gas, the equation of state is:

$$PV = RT \quad (5)$$

Writing the ideal gas law for states 1 and 2: $P_1 V_1 = RT_1$

$$P_2 V_2 = RT_2$$

Solving each equation for volume: $V_1 = \frac{RT_1}{P_1}$

$$V_2 = \frac{RT_2}{P_2}$$

Substituting the volume expressions into the pressure ratio relation:

$$\left(\frac{P_2}{P_1}\right)^{1/\gamma} = \frac{(RT_1/P_1)}{(RT_2/P_2)} \quad (6)$$

Canceling R :

$$\left(\frac{P_2}{P_1}\right)^{1/\gamma} = \frac{T_1 P_2}{T_2 P_1} \quad (7)$$

Rearranging:

$$\frac{T_2}{T_1} = \frac{P_2}{P_1} \left(\frac{P_1}{P_2}\right)^{1/\gamma} \quad (8)$$

Combining the pressure terms:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{1-1/\gamma} \quad (9)$$

Recognizing that:

$$1 - \frac{1}{\gamma} = \frac{\gamma - 1}{\gamma} \quad (10)$$

The final isentropic temperature–pressure relation becomes:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{(\gamma-1)/\gamma} \quad (11)$$

Defining the compressor pressure ratio:

$$r_p = \frac{P_2}{P_1} \quad (12)$$

The temperature relation can be written as:

$$T_2 = T_1 r_p^{(\gamma-1)/\gamma} \quad (13)$$

For the turbine expansion process from state 3 to state 4:

$$P_3 V_3^\gamma = P_4 V_4^\gamma \quad (14)$$

Following the same steps:

$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3}\right)^{(\gamma-1)/\gamma} \quad (15)$$

Since $P_4 = P_1$ and $P_3 = P_2$:

$$\frac{T_4}{T_3} = \left(\frac{1}{r_p}\right)^{(\gamma-1)/\gamma} \quad (16)$$

Thus, the turbine exit temperature is:

$$T_4 = T_3 r_p^{-(\gamma-1)/\gamma} \quad (17)$$

2.2.2 General Steady-State Energy Balance

The most general steady-state energy balance for a control volume is:

$$\frac{dE_{CV}}{dt} = \dot{Q} - \dot{W} + \sum_{in} \dot{n}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) - \sum_{out} \dot{n}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) \quad (18)$$

For steady-state operation:

$$\frac{dE_{CV}}{dt} = 0 \quad (19)$$

Thus:

$$0 = \dot{Q} - \dot{W} + \sum_{in} \dot{n}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) - \sum_{out} \dot{n}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) \quad (20)$$

2.2.3 Energy Balance for the Compressor

For the compressor, there is one inlet (state 1) and one outlet (state 2):

$$0 = \dot{Q}_c - \dot{W}_c + \dot{n} \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) - \dot{n} \left(h_2 + \frac{V_2^2}{2} + gz_2 \right) \quad (21)$$

Assuming the compressor is adiabatic:

$$\dot{Q}_c = 0 \quad (22)$$

Assuming negligible kinetic and potential energy changes:

$$\frac{V_1^2}{2} \approx \frac{V_2^2}{2}, \quad gz_1 \approx gz_2 \quad (23)$$

The energy balance reduces to:

$$0 = -\dot{W}_c + \dot{n}(h_1 - h_2) \quad (24)$$

Rearranging:

$$\dot{W}_c = \dot{n}(h_1 - h_2) \quad (25)$$

Dividing by the molar flow rate:

$$w_c = h_1 - h_2 \quad (26)$$

Using the ideal gas relation $h = C_p T$ which is $h_2 - h_1 = C_p(T_2 - T_1)$ and rearranging it $h_1 - h_2 = -C_p(T_2 - T_1)$

$$w_c = -C_p(T_2 - T_1) \quad (27)$$

Substituting the isentropic relation:

$$T_2 = T_1 r_p^{(\gamma-1)/\gamma} \quad (28)$$

The compressor work becomes:

$$w_c = -C_p \left(T_1 r_p^{(\gamma-1)/\gamma} - T_1 \right) \quad (29)$$

2.2.4 Energy Balance for the Turbine

For the turbine, there is one inlet (state 3) and one outlet (state 4):

$$0 = \dot{Q}_t - \dot{W}_t + \dot{n} \left(h_3 + \frac{V_3^2}{2} + gz_3 \right) - \dot{n} \left(h_4 + \frac{V_4^2}{2} + gz_4 \right) \quad (30)$$

Assuming the turbine is adiabatic:

$$\dot{Q}_t = 0 \quad (31)$$

Assuming negligible kinetic and potential energy changes:

$$\frac{V_3^2}{2} \approx \frac{V_4^2}{2}, \quad gz_3 \approx gz_4 \quad (32)$$

The energy balance reduces to:

$$0 = -\dot{W}_t + \dot{n}(h_3 - h_4) \quad (33)$$

Rearranging:

$$\dot{W}_t = \dot{n}(h_3 - h_4) \quad (34)$$

Dividing by the molar flow rate:

$$w_t = h_3 - h_4 \quad (35)$$

Using the ideal gas relation $h = C_p T$ which is $h_4 - h_3 = C_p(T_4 - T_3)$ and rearranging it $h_3 - h_4 = -C_p(T_4 - T_3)$:

$$w_t = -C_p(T_4 - T_3) \quad (36)$$

Substituting the isentropic relation:

$$T_4 = T_3 r_p^{-(\gamma-1)/\gamma} \quad (37)$$

The turbine work becomes:

$$w_t = -C_p \left(T_3 r_p^{-(\gamma-1)/\gamma} - T_3 \right) \quad (38)$$

2.2.5 Net Work Expression

$$w_{net} = w_t - w_c \quad (39)$$

Substitute full expressions:

$$w_{net} = -C_p(T_3 r_p^{-(\gamma-1)/\gamma} - T_3) - C_p(T_1 r_p^{(\gamma-1)/\gamma} - T_1) \quad (40)$$

Factor $-C_p$:

$$w_{net} = -C_p \left[T_3 r_p^{-(\gamma-1)/\gamma} - T_3 + T_1 r_p^{(\gamma-1)/\gamma} - T_1 \right] \quad (41)$$

2.2.6 Taking the Derivative

Take the derivative of w_{net} with respect to r_p :

$$\frac{dw_{net}}{dr_p} = -C_p \left[T_3 \left(-\frac{\gamma-1}{\gamma} \right) r_p^{-(\gamma-1)/\gamma-1} + T_1 \left(\frac{\gamma-1}{\gamma} \right) r_p^{((\gamma-1)/\gamma)-1} \right] \quad (42)$$

Set derivative equal to zero:

$$\frac{dw_{net}}{dr_p} = 0 \quad (43)$$

Rearranging:

$$T_3 r_p^{-(\gamma-1)/\gamma} = T_1 r_p^{(\gamma-1)/\gamma} \quad (44)$$

Multiply both sides:

$$r_p^{2(\gamma-1)/\gamma} = \frac{T_3}{T_1} \quad (45)$$

Solve for r_p :

$$r_p = \left(\frac{T_3}{T_1} \right)^{\frac{\gamma}{2(\gamma-1)}} \quad (46)$$

Substitute values:

$$r_p = \left(\frac{1400}{300} \right)^{\frac{1.4}{0.8}} = 14.82 \quad (47)$$

2.3 Excel Solver Method

All given data were entered into Excel. T_2 was calculated using $T_2 = T_1 r_p^{(\gamma-1)/\gamma}$ and T_4 was calculated using $T_4 = T_3 r_p^{-(\gamma-1)/\gamma}$. The net work expression

$$w_{net} = C_p [(T_3 - T_4) - (T_2 - T_1)]$$

was used. Excel Solver was set to maximize w_{net} by changing r_p , which resulted in $r_p = 14.82$.

A-1b				
Data Given				
P1	100			
T1	300			
T3	1400			
Cp	29.1			
Gamma	1.4			
Exponent	0.28571429			
Pressure Ratio	T2	T4	Wnet	
14.81705443	648.074074	648.0741	11752.09	

Figure 2: Excel Image of Sheet After the Solver Function

2.4 Finding T_2 , T_4 , and P_2 :

$$\frac{T_2}{T_1} = r^{\frac{\gamma-1}{\gamma}} \quad \frac{T_4}{T_3} = r^{-\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{T_1} = x \quad \frac{T_4}{T_3} = x$$

$$\frac{T_2}{T_1} = r^{\frac{\gamma-1}{\gamma}} \times \frac{T_4}{T_3} = r^{-\frac{\gamma-1}{\gamma}}$$

$$\frac{T_2}{T_1} \times \frac{T_4}{T_3} = r^0$$

$$\frac{T_2}{T_1} \times \frac{T_4}{T_3} = 1$$

Cross Product

$$T_2 = T_3 = 1400K \quad \text{and} \quad T_1 = T_4 = 300K$$

$$\frac{P_2}{P_1} = r_p = 14.82bar$$

$$\frac{P_2}{1bar} = 14.82bar$$

$$P_2 = 14.82bar$$

2.5 Finding the thermal efficiency:

$$\frac{\dot{w}}{\dot{m}} = -C_p \left(T_1 r_p^{\frac{\gamma-1}{\gamma}} - T_1 \right) - C_p \left(T_3 r_p^{-\frac{\gamma-1}{\gamma}} - T_3 \right)$$

$$\frac{\dot{w}}{\dot{m}} = -29.1 \frac{J}{mol \cdot K} \left(300K(14.82)^{\frac{1.4-1}{1.4}} - 300 \right) - 29.1 \frac{J}{mol \cdot K} \left(1400K(14.82)^{-\frac{1.4-1}{1.4}} - 1400K \right)$$

$$\frac{\dot{w}}{\dot{m}} = 11,752.09 \text{ J/mol}$$

$$Q = mC_p \Delta T (T_3 - T_2)$$

only combustor is accepting heat

$$\frac{Q}{m} = 29.1 \frac{J}{mol \cdot K} \left(1400K - T_1 r_p^{\frac{\gamma-1}{\gamma}} \right)$$

$$\frac{Q}{m} = 29.1 \frac{J}{mol \cdot K} \left(1400K - 300(14.82)^{\frac{1.4-1}{1.4}} \right)$$

$$\frac{Q}{m} = 21,879.97 \text{ J/mol}$$

$$\eta = \frac{W_{net}}{Q_H} = \frac{11,752.09 \text{ J/mol}}{21,879.97 \text{ J/mol}} = 0.5371 \times 100\%$$

$$\eta = 53.71\%$$

2.6 Finding the Carnot Efficiency

$$\eta_{\max} = 1 - \frac{T_C}{T_H} = 1 - \frac{300 \text{ K}}{1400 \text{ K}} = 0.7857 \times 100 = 78.57\%$$

2.7 Calculating net power at maximum efficiency:

$$\begin{aligned}\dot{w} &= \dot{m} \left[-C_p (T_2 - T_1) - C_p (T_4 - T_3) \right] \\ \dot{w} &= \dot{m} \left[-29.1 \frac{\text{J}}{\text{mol} \cdot \text{K}} (1400 \text{ K} - 300 \text{ K}) - 29.1 \frac{\text{J}}{\text{mol} \cdot \text{K}} (300 \text{ K} - 1400 \text{ K}) \right] \\ \dot{w} &= 0 \text{ J/mol}\end{aligned}$$

2.8 Minimum Air Flow Rate:

Air Flow Rate = Molar Flow Rate

$$\dot{n} = \frac{\text{mol}}{\text{s}}$$

$$W_{\text{net}} = 11,752.09 \frac{\text{J}}{\text{mol}}$$

$$50 \text{ MW} \times \frac{10^6 \text{ W}}{1 \text{ MW}} \times \frac{1 \text{ J/s}}{1 \text{ W}} = 50 \times 10^6 \frac{\text{J}}{\text{s}}$$

$$\dot{n} = \left(50 \times 10^6 \frac{\text{J}}{\text{s}} \right) \left(\frac{\text{mol}}{11,752.09 \text{ J}} \right) = 4254.56 \frac{\text{mol}}{\text{s}}$$

1 mol $\approx$ 0.8366 scf

1 Day = 86400 Seconds

To Convert to MMSCFD:

$$\text{Minimum air flow rate} = 4254.56 \frac{\text{mol}}{\text{s}} \left(\frac{0.8366 \text{ scf}}{1 \text{ mol}} \right) \left(\frac{86400 \text{ s}}{1 \text{ Day}} \right) \left(\frac{1 \text{ MMSCFD}}{10^6 \text{ scf}} \right) \approx 307.53 \text{ MMSCFD}$$

3 Problem B: Numerical Solution of an Algebraic Equation

3.1 Finding the Derivative:

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i (k_i - 1)}{1 + \alpha (k_i - 1)} = 0$$

$$F'(\alpha) = \sum_{i=1}^{14} \frac{(0) (1 + \alpha (k_i - 1)) - z_i (k_i - 1) (0 + (k_i - 1))}{(1 + \alpha (k_i - 1))^2}$$

$$F'(\alpha) = \sum_{i=1}^{14} \frac{-z_i (k_i - 1)^2}{(1 + \alpha (k_i - 1))^2}$$

3.2 Calculating the Vapor fraction α :

- By using the data given below (leave space for the table) and by starting at $\alpha = 0.5$

z(1)	0.117598	K(1)	49.905777
z(2)	0.002775	K(2)	13.445753
z(3)	0.003435	K(3)	3.4749140
z(4)	0.003171	K(4)	1.2825470
z(5)	0.001189	K(5)	0.6281980
z(6)	0.000793	K(6)	0.4776770
z(7)	0.033971	K(7)	0.0644180
z(8)	0.044667	K(8)	0.0555770
z(9)	0.002572	K(9)	0.2268650
z(10)	0.065649	K(10)	0.1835740
z(11)	0.412276	K(11)	0.0927070
z(12)	0.032581	K(12)	0.1367150
z(13)	0.225623	K(13)	0.0722520
z(14)	0.053700	K(14)	0.0382340

3.2.1 By Using Newton method:

First Set a table as shown below and then solve each cell , for example $F(0.5)$ will be the Sum product of $z_i(k_i - 1)/(1 + \alpha(k_i - 1))$ and $F'(0.5)$ the Sum product of $-z_i(k_i - 1)^2/(1 + \alpha(k_i - 1))^2$. The next α_i is given by

$$\alpha_i - \frac{F(\alpha_i)}{F'(\alpha_i)}$$

Solving for the root through five iterations, the value of α was approximated to be:

$$\alpha \approx 0.1155$$

	A	B	C	D	E	F	G	H	I
1	B- Newton's Method								
2									
3	alpha	F(vapor Fraction)	F'(Vapor Fraction)	New alpha		Z	K	K-1	(K-1)^2
4	0.5000	-1.2233	-2.8965	0.0777		0.117598	49.90578	48.90578	2391.775
5	0.0777	0.3701	-13.1785	0.1058		0.002775	13.44575	12.44575	154.8968
6	0.1058	0.0762	-8.3624	0.1149		0.003435	3.474914	2.474914	6.125199
7	0.1149	0.0045	-7.4106	0.1155		0.003171	1.282547	0.282547	0.079833
8	0.1155	0.0000	-7.3541	0.1155		0.001189	0.628198	-0.3718	0.138237
9	0.1155	0.0000	-7.3539	0.1155		0.000793	0.477677	-0.52232	0.272821
10	0.1155	0.0000	-7.3539	0.1155		0.033971	0.064418	-0.93558	0.875314
11	0.1155	0.0000	-7.3539	0.1155		0.044667	0.055577	-0.94442	0.891935
12	0.1155	0.0000	-7.3539	0.1155		0.002572	0.226865	-0.77314	0.597738
13	0.1155	0.0000	-7.3539	0.1155		0.065649	0.183574	-0.81643	0.666551
14						0.412276	0.092707	-0.90729	0.823181
15						0.032581	0.136715	-0.86329	0.745261
16						0.225623	0.072252	-0.92775	0.860716
17						0.0537	0.038234	-0.96177	0.924994
18									

3.2.2 By using secant method :

Set a table as shown below and then solve each cell as follows :
So we will start at $\alpha_0 = 0.5$ and $\alpha_1 = 1$ then solve

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i(k_i - 1)}{1 + \alpha(k_i - 1)}$$

for each α and then to solve for the new α you do the following :

$$\alpha_2 \approx \frac{\alpha_0 F(\alpha_1) - \alpha_1 F(\alpha_0)}{F(\alpha_1) - F(\alpha_0)}$$

which is the new α and solve for α_2 . Then start again at α_0 the Second row where we replace it with α_1 from the first row and replace α_1 in the second row with α_2 that was Calculated in row one. After 14 iterations which that also gave us

$$\alpha \approx 0.1155$$

19							
20	Secant Method						
21	alpha 0	alpha 1	F(alpha 0)	F(alpha 1)	alpha 2	F(alpha 2)	
22	0.5000	1.0000	-1.2233	-9.9222	0.4297	-1.0330	
23	1.0000	0.4297	-9.9222	-1.0330	0.3634	-0.8687	
24	0.4297	0.3634	-1.0330	-0.8687	0.0130	2.7554	
25	0.3634	0.0130	-0.8687	2.7554	0.2794	-0.6589	
26	0.0130	0.2794	2.7554	-0.6589	0.2280	-0.5130	
27	0.2794	0.2280	-0.6589	-0.5130	0.0472	0.9405	
28	0.2280	0.0472	-0.5130	0.9405	0.1642	-0.2779	
29	0.0472	0.1642	0.9405	-0.2779	0.1375	-0.1426	
30	0.1642	0.1375	-0.2779	-0.1426	0.1093	0.0469	
31	0.1375	0.1093	-0.1426	0.0469	0.1163	-0.0061	
32	0.1093	0.1163	0.0469	-0.0061	0.1155	-0.0002	
33	0.1163	0.1155	-0.0061	-0.0002	0.1155	0.0000	
34	0.1155	0.1155	-0.0002	0.0000	0.1155	0.0000	
35	0.1155	0.1155	0.0000	0.0000	0.1155	0.0000	
36	0.1155	0.1155	0.0000	0.0000	0.1155	0.0000	
37							
38							

3.3 By Using Goal seek:

Set α to 0.5 and solve for $F(\alpha)$ then use the function goal seek in Excel to set $F(\alpha) = 0$ by changing α which also gave us

$$\alpha \approx 0.1155$$

39			
40	Goal Seek		
41	alpha	F(alpha)	
42	0.1155	0.0000	
43			
44			

4 Conclusion

All thermodynamic relations and numerical methods were written and solved without simplification. The Brayton cycle optimization was derived analytically and verified numerically. Numerical methods for vapor fraction calculations produced consistent results, demonstrating the importance of step-by-step analysis in chemical engineering thermodynamics.

In this assignment, the ideal Brayton cycle was analyzed using basic thermodynamic principles and steady-state energy balances. The isentropic relations and energy balance equations were used to find the pressure ratio that maximizes the net power output, and this result was confirmed using Excel Solver. From the derived equations and given data, the thermal efficiency, Carnot efficiency, net power, and minimum air flow were found. This showed how this problem depends heavily on the pressure ratio and temperature of the system. The temperature being so drastically different at maximum efficiency shows this relationship best. The Carnot efficiency also showed the theoretical limit of performance for a heat engine operating between the same temperatures.

In Problem B, numerical methods were applied to solve the nonlinear equation for the vapor fraction. The derivative was found analytically, and the Newton method, Secant method, and Excel Goal Seek all converged to the same value for the vapor fraction, which confirmed the solution. This showed the importance of numerical methods when analytical solutions are not efficient.